String diagrams

bpaul :3

March 19, 2026

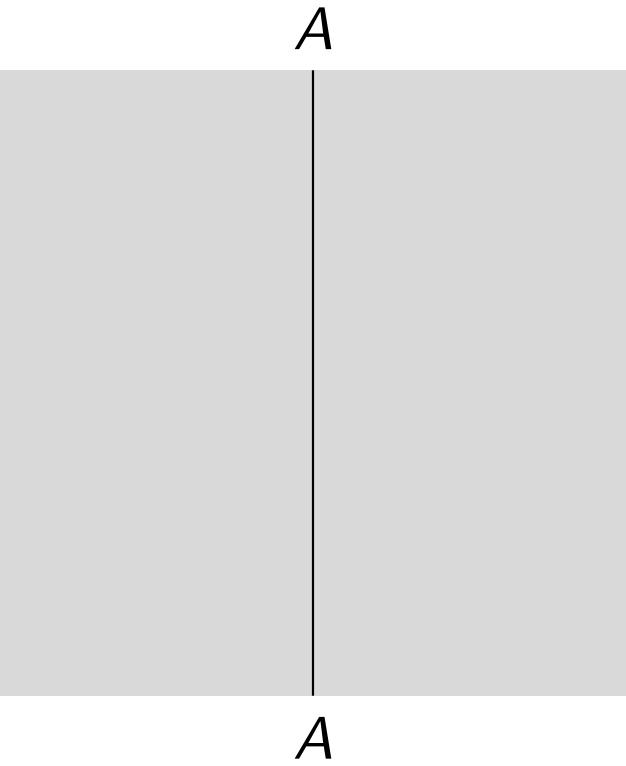
Outline

The calculus of string diagrams

Worked example: monoids

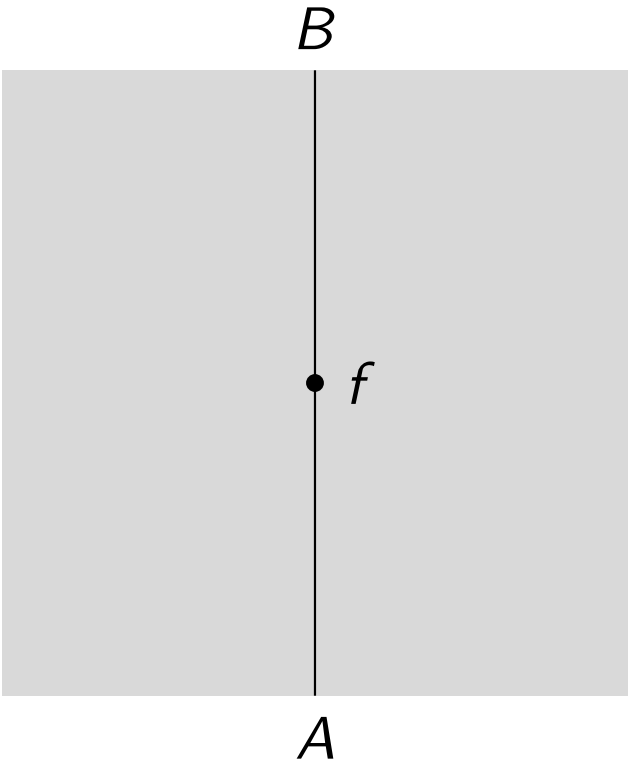
Our first diagrams

Our first diagrams



Here A is a set.

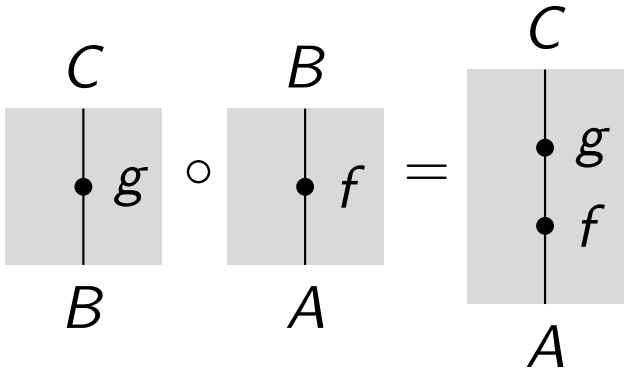
Our first diagrams



Here A and B are sets, and $f : A \to B$ is a function.

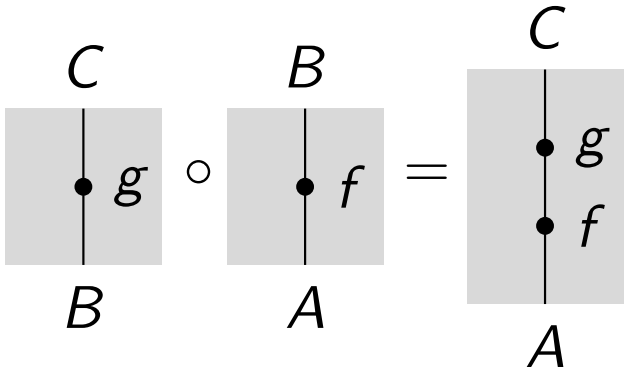
Compositions

Vertical composition $a \mapsto g(f(a))$:

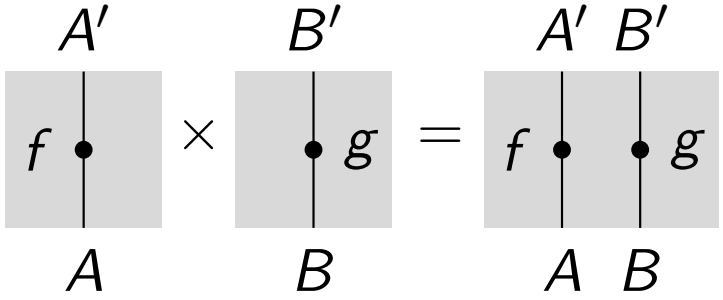


Compositions

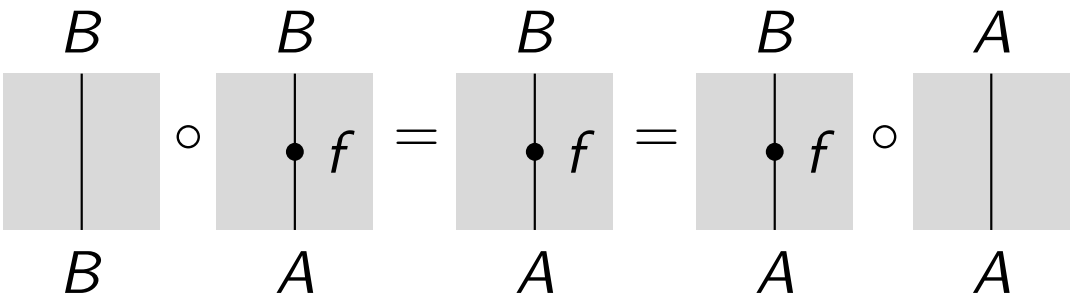
Vertical composition $a \mapsto g(f(a))$:

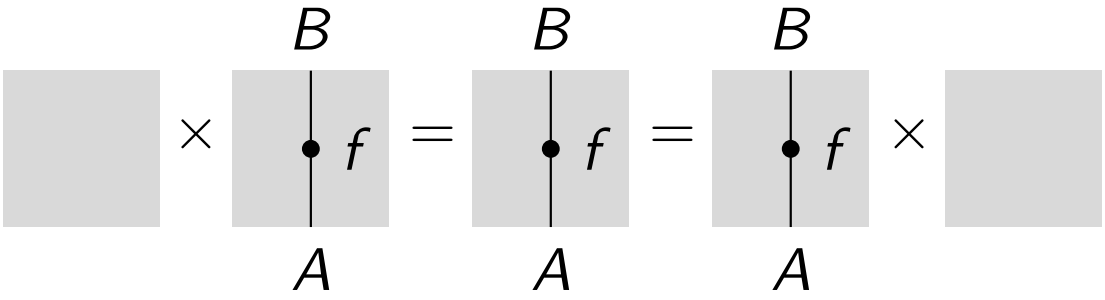


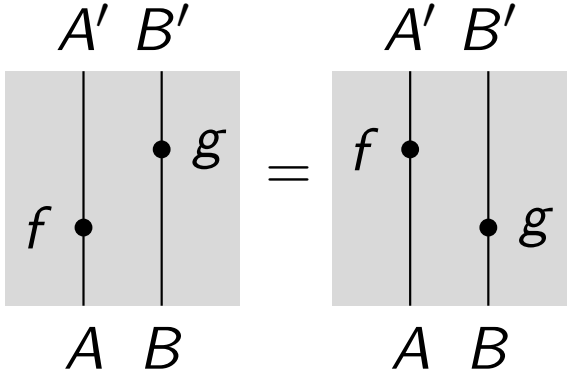
Horizontal composition $(a, b) \mapsto (f(a), g(b))$:



Rules

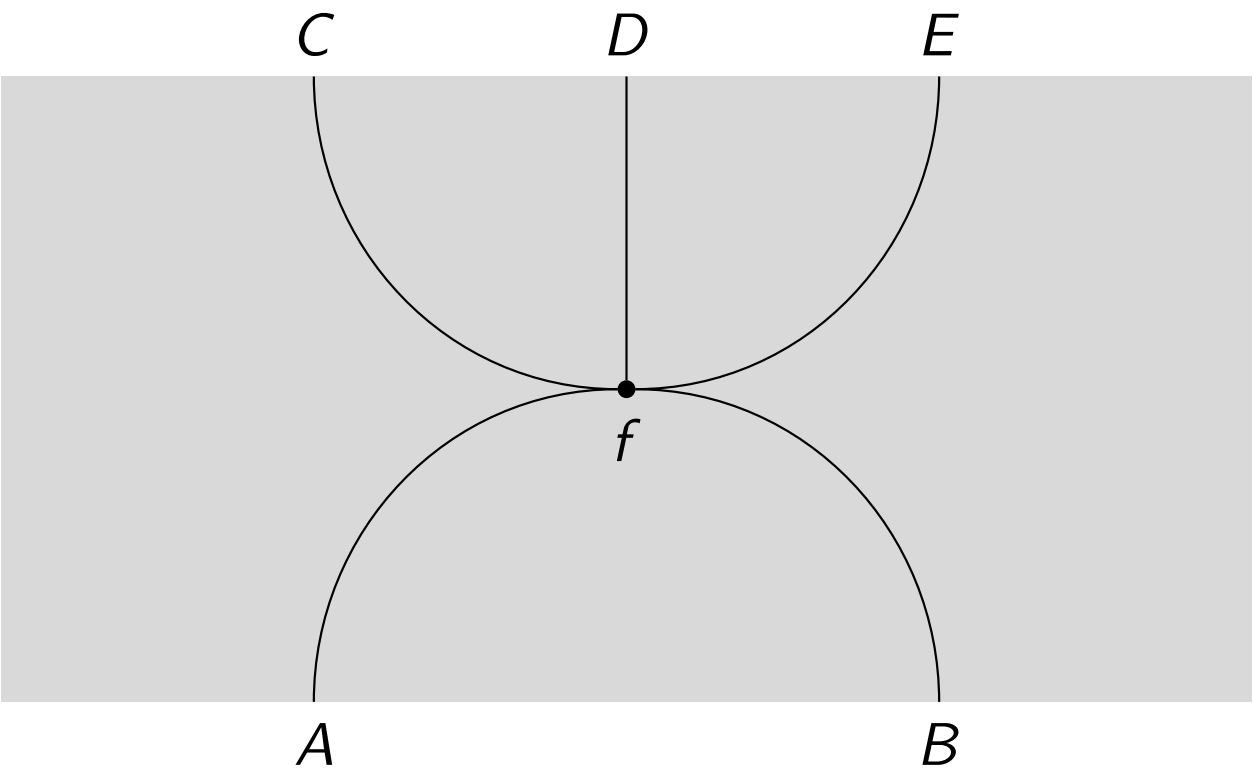




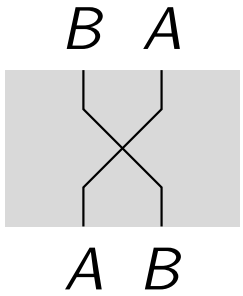


Multivalued functions

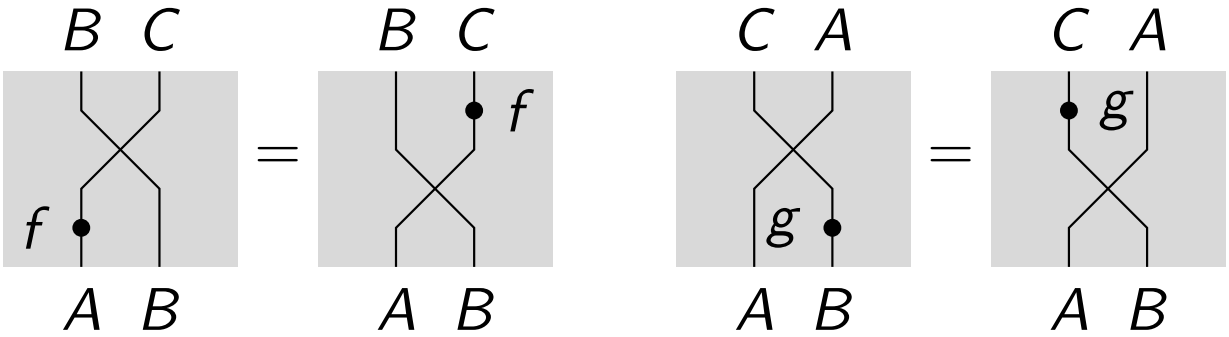
If $f : A \times B \to C \times D \times E$ then we can draw it like so:



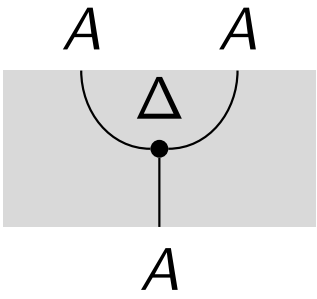
More constructs

Crossing over $(a, b) \mapsto (b, a)$: 

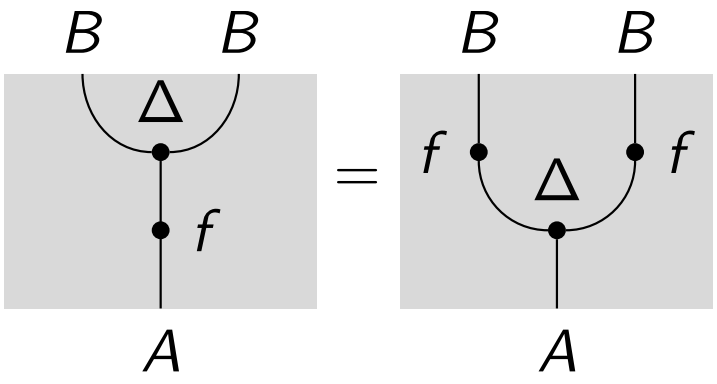
, satisfying:

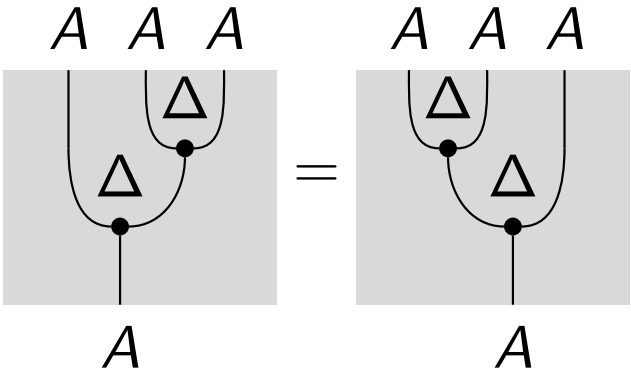


More constructs

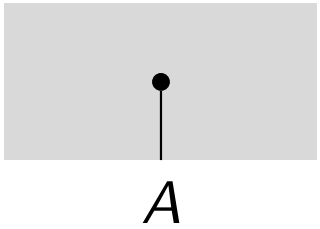
Duplicating elements $a \mapsto (a, a)$: 

, satisfying:

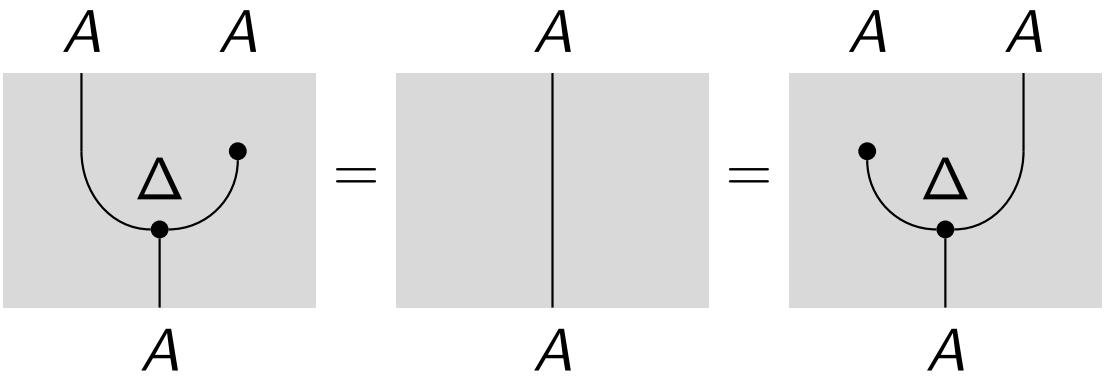




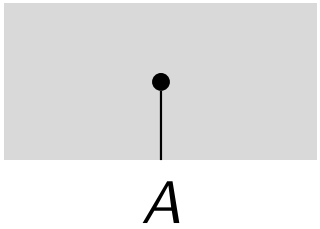
More constructs

Deleting elements: 

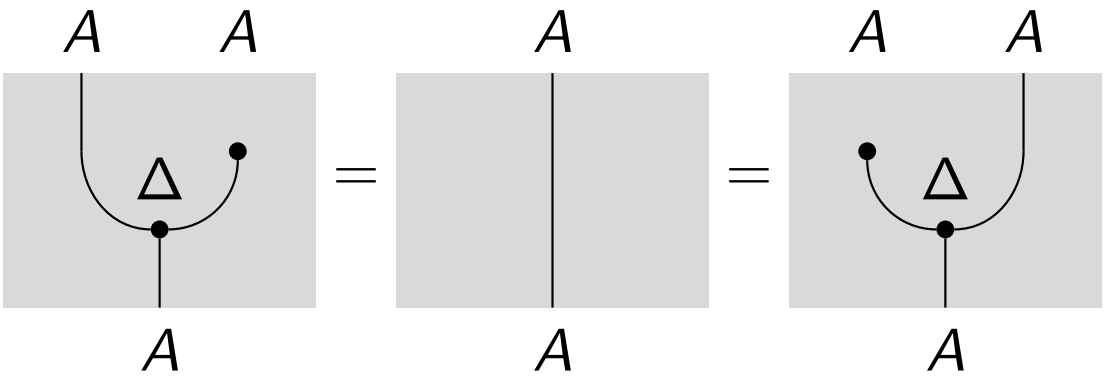
, satisfying:



More constructs

Deleting elements: 

, satisfying:



etc.

Monoids

Definition

A monoid is a set M equipped with a multiplication $\mu : M \times M \to M$ (we write $\mu(a, b) = a * b$) and an element $e \in M$ called the identity satisfying:

- (associativity) for any $a, b, c \in M$, $a * (b * c) = (a * b) * c$,
- (identity) for any $a \in M$, $e * a = a = a * e$.

Monoids

Definition

A monoid is a set M equipped with a multiplication $\mu : M \times M \to M$ (we write $\mu(a, b) = a * b$) and an element $e \in M$ called the identity satisfying:

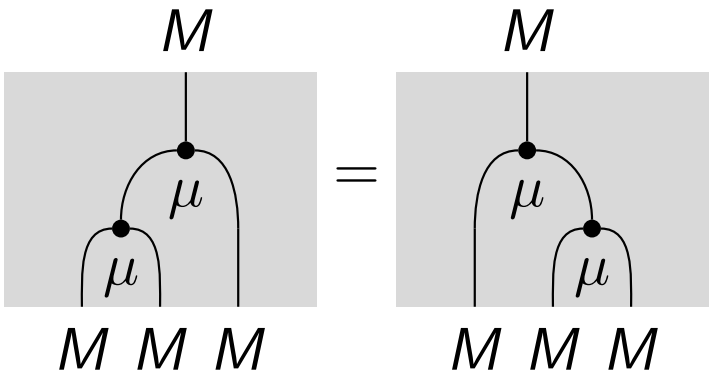
- (associativity) for any $a, b, c \in M$, $a * (b * c) = (a * b) * c$,
- (identity) for any $a \in M$, $e * a = a = a * e$.

Example

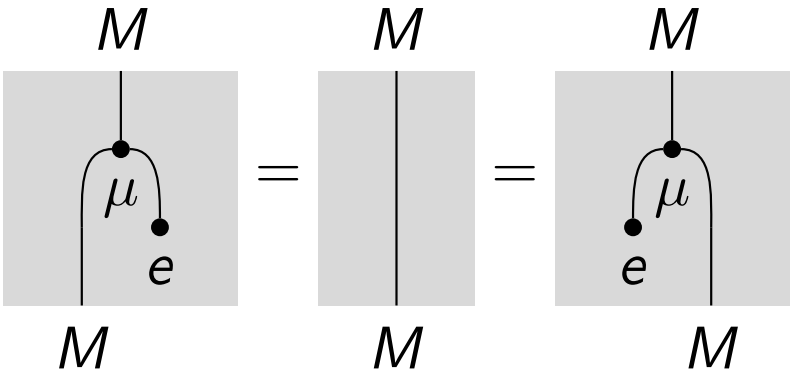
- The natural numbers under addition $(\mathbb{N}, +, 0)$
- The natural numbers under multiplication $(\mathbb{N}, *, 1)$
- Matrices under multiplication $(\mathrm{Mat}_n, *, I_n)$
- Strings under concatenation $(A^*, ||,)$

Monoids diagramatically

The map $\mu : M \times M \to M$ and element $e \in M$ satisfy:
Associativity:



Identity:

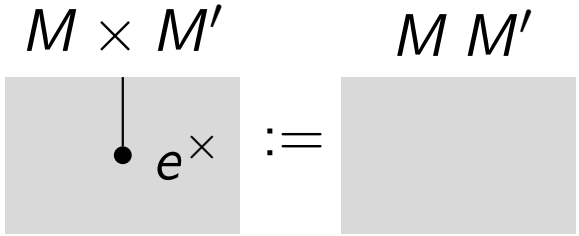


Product monoid

Let $(M, \mu, e), (M', \mu', e')$ be monoids. Define the product monoid, with set $M \times M'$ as follows.

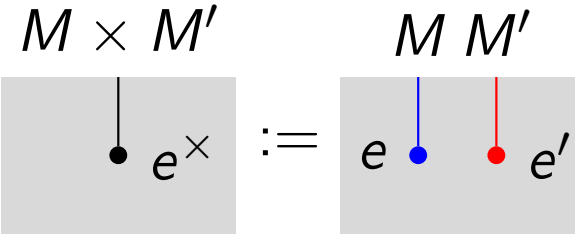
Product monoid

Let $(M, \mu, e), (M', \mu', e')$ be monoids. Define the product monoid, with set $M \times M'$ as follows.
Identity:



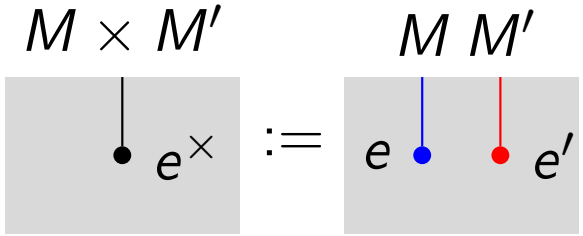
Product monoid

Let $(M, \mu, e), (M', \mu', e')$ be monoids. Define the product monoid, with set $M \times M'$ as follows.
Identity:

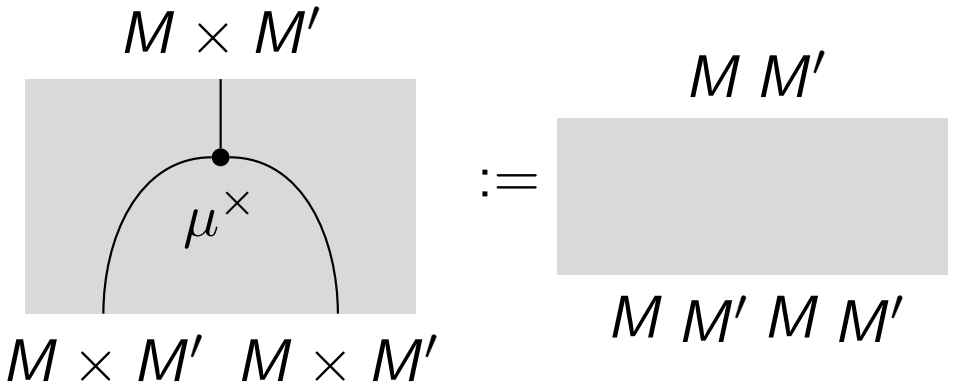


Product monoid

Let $(M, \mu, e), (M', \mu', e')$ be monoids. Define the product monoid, with set $M \times M'$ as follows.
Identity:

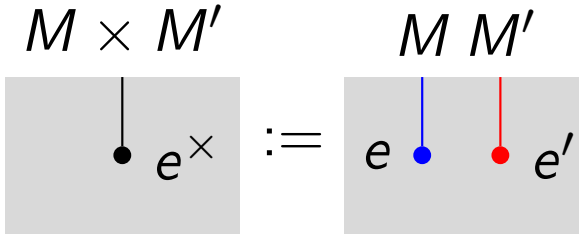


Multiplication:

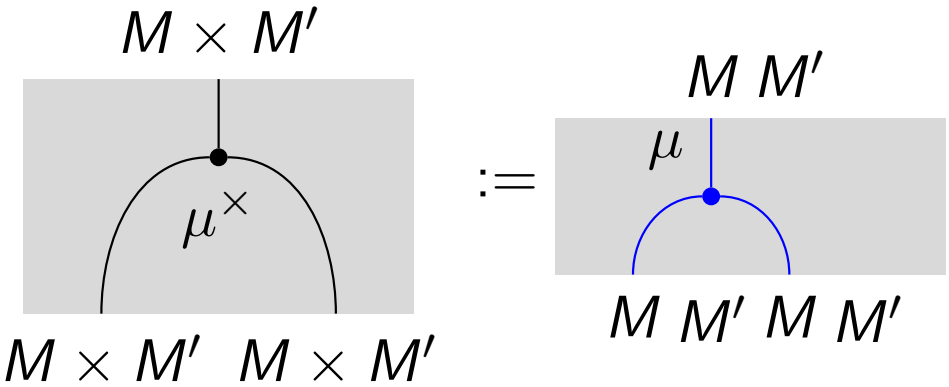


Product monoid

Let $(M, \mu, e), (M', \mu', e')$ be monoids. Define the product monoid, with set $M \times M'$ as follows.
Identity:

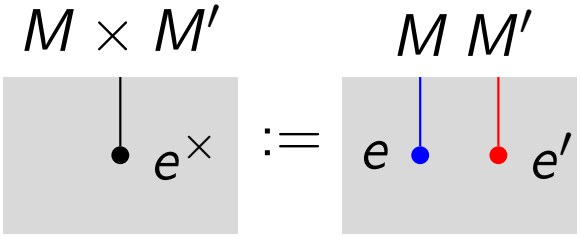


Multiplication:

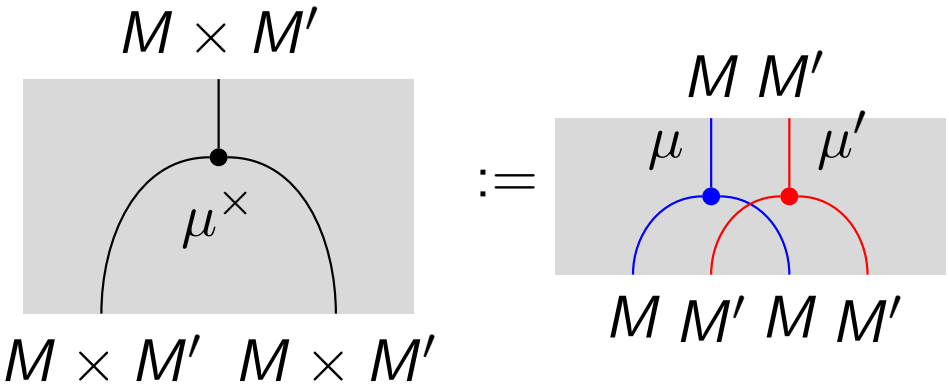


Product monoid

Let $(M, \mu, e), (M', \mu', e')$ be monoids. Define the product monoid, with set $M \times M'$ as follows.
Identity:

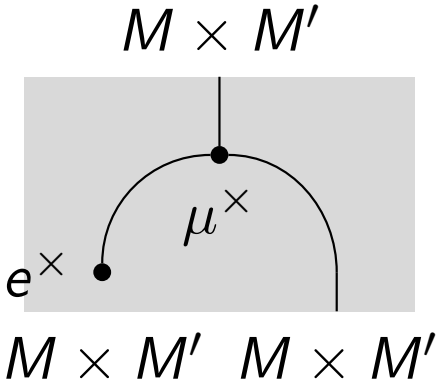


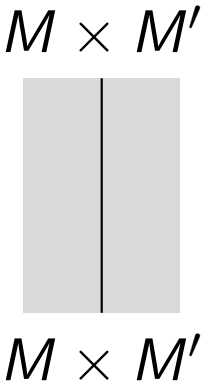
Multiplication:



Product monoid

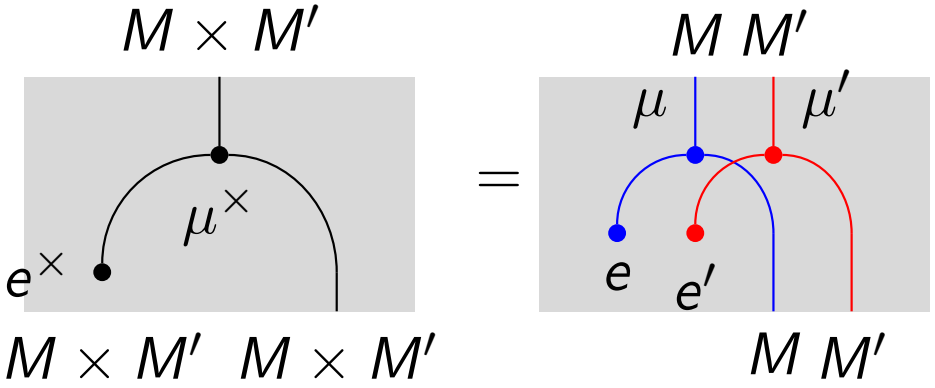
Left identity:

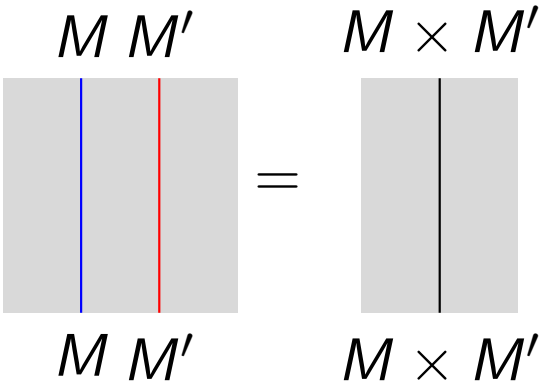




Product monoid

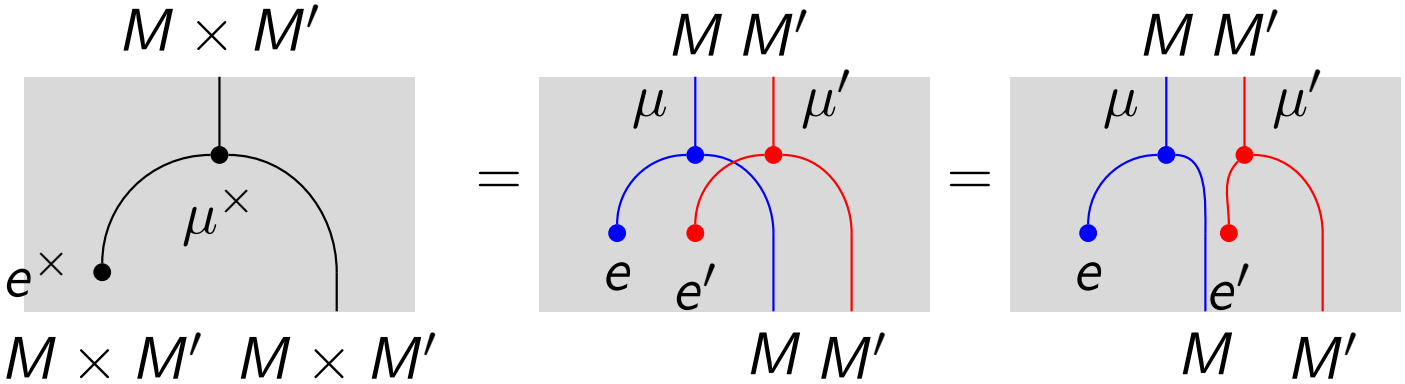
Left identity:

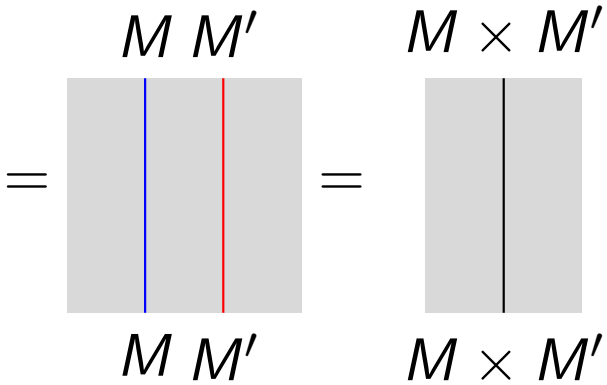




Product monoid

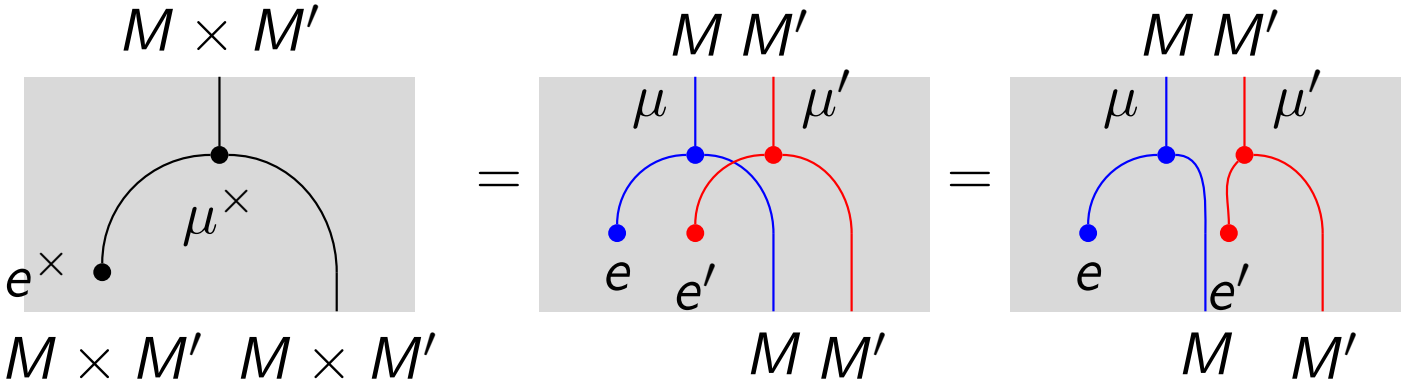
Left identity:

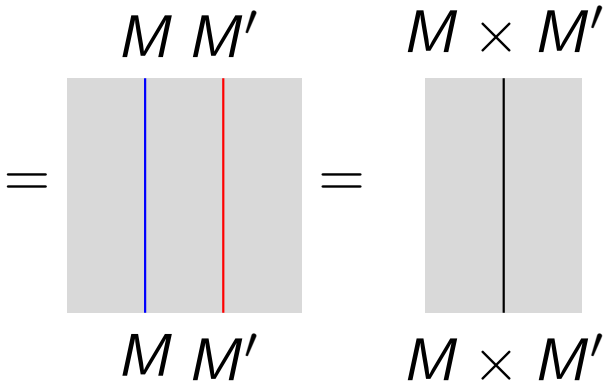




Product monoid

Left identity:

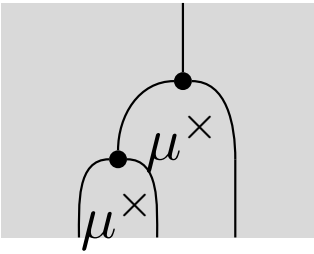


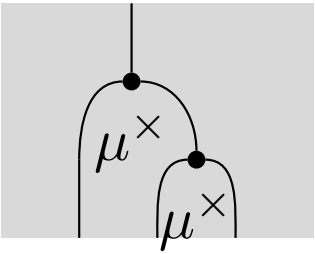


Right identity is similar.

Product monoid

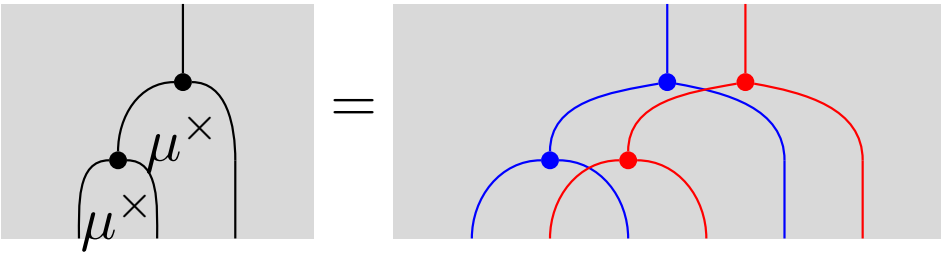
Associativity:

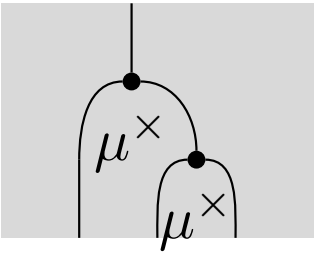




Product monoid

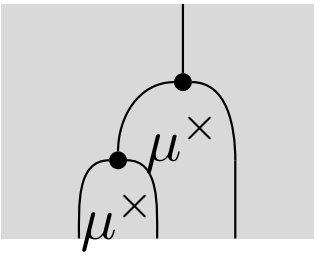
Associativity:



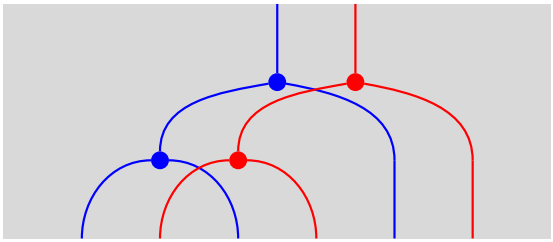


Product monoid

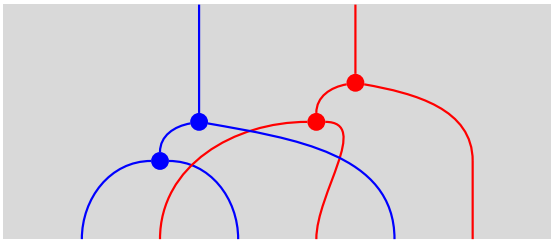
Associativity:

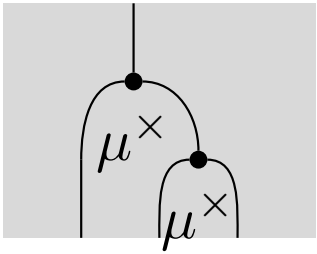


$=$



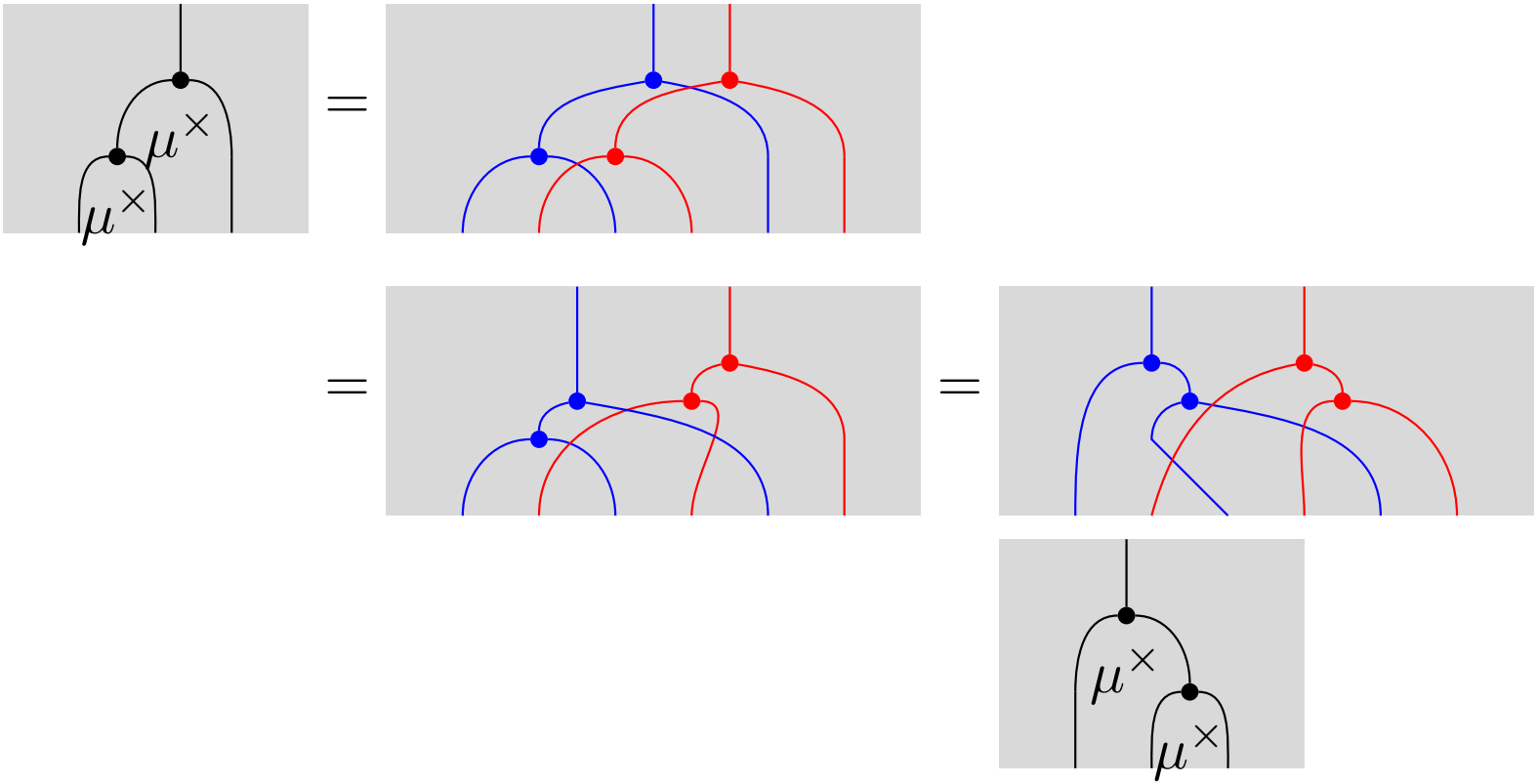
$=$





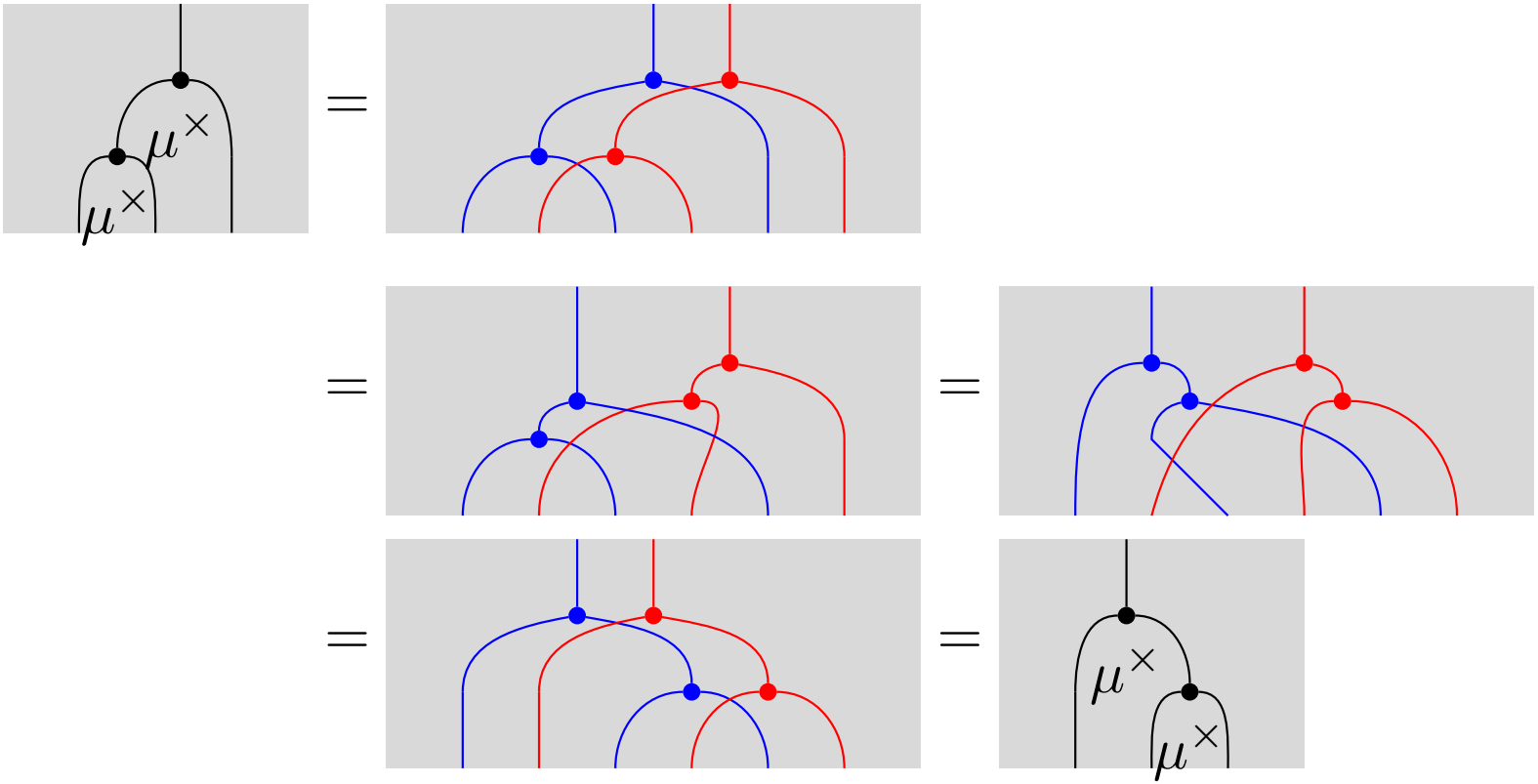
Product monoid

Associativity:



Product monoid

Associativity:



Monoid homomorphisms

Definition

Let $(M, \mu, e), (M', \mu', e')$ be monoids. A function $f : M \to M'$ is a monoid homomorphism if

- for any $a, b \in M$, $f(a * b) = f(a) *' f(b)$,
- and $f(e) = e'$.

Monoid homomorphisms

Definition

Let $(M, \mu, e), (M', \mu', e')$ be monoids. A function $f : M \to M'$ is a monoid homomorphism if

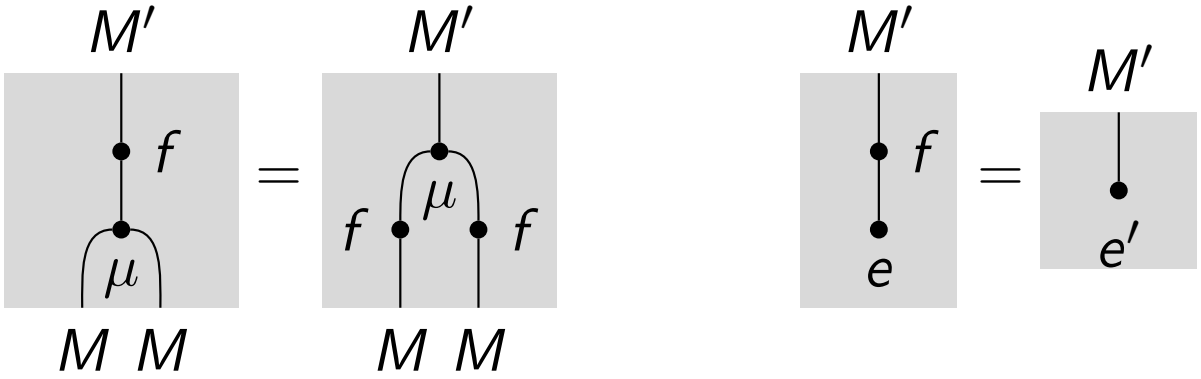
- for any $a, b \in M$, $f(a * b) = f(a) *' f(b)$,
- and $f(e) = e'$.

Example

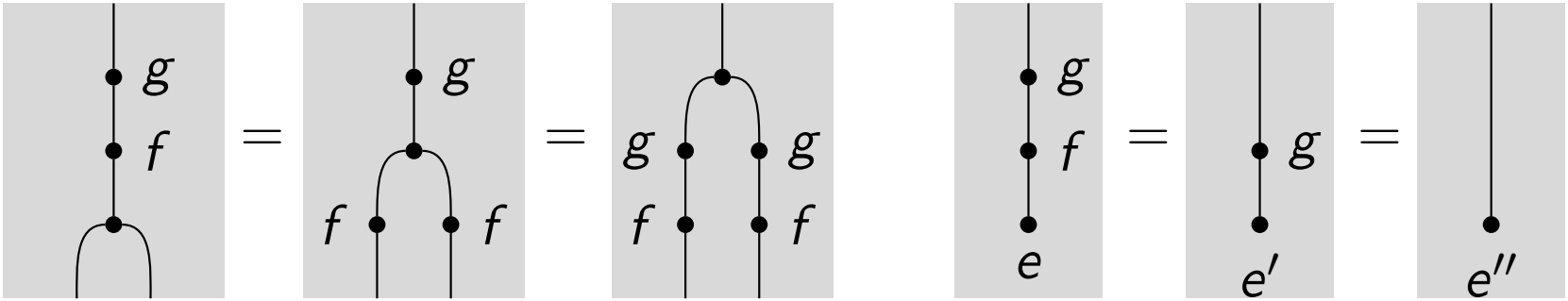
- From $(\mathbb{N}, +, 0)$ to $(\mathbb{N}, *, 1)$, map $n \mapsto 2^n$.
- From $(\mathbb{R}, *, 0)$ to $(\mathrm{Mat}_n, *, I_n)$, map $x \mapsto xI_n$.
- From $(\mathbb{Z}_4, +, 0)$ to $(S^1, *, 1)$, send $n \mapsto e^{\frac{n\pi i}{2}}$.

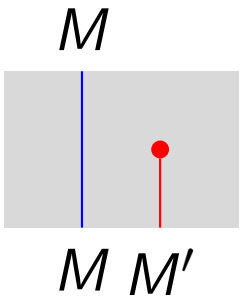
Homomorphisms diagramatically

A function $f : M \to M'$ is a homomorphism if:

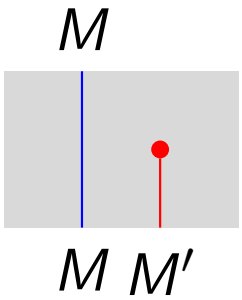


The composite of two homomorphisms is a homomorphism:

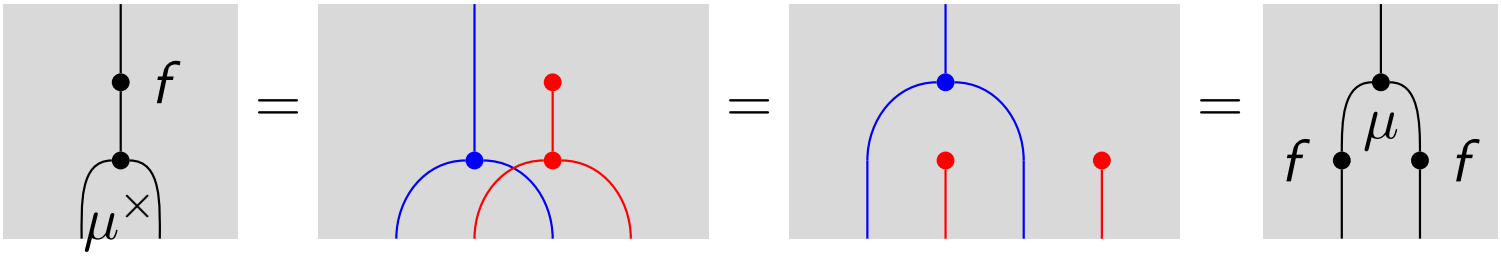


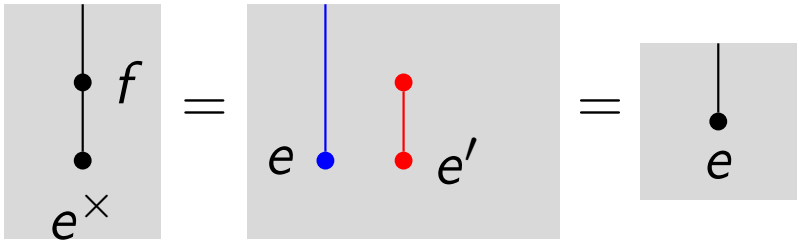
Is $f =$ 

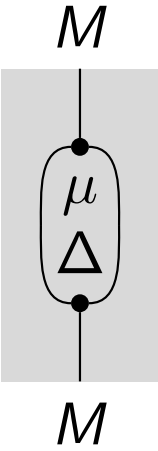
a homomorphism?

Is $f =$ 

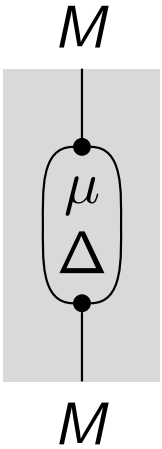
a homomorphism? Yes!





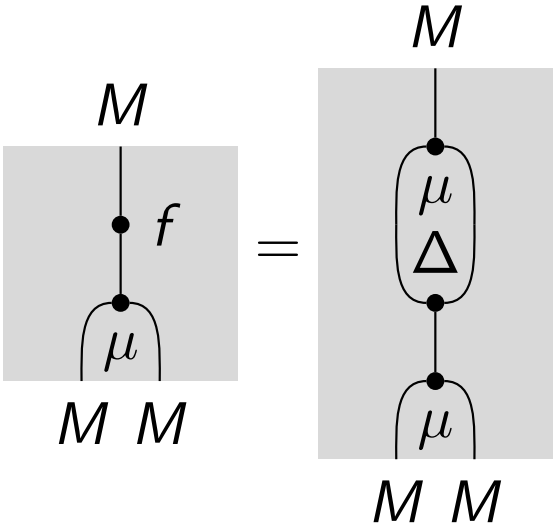
Is $f =$ 

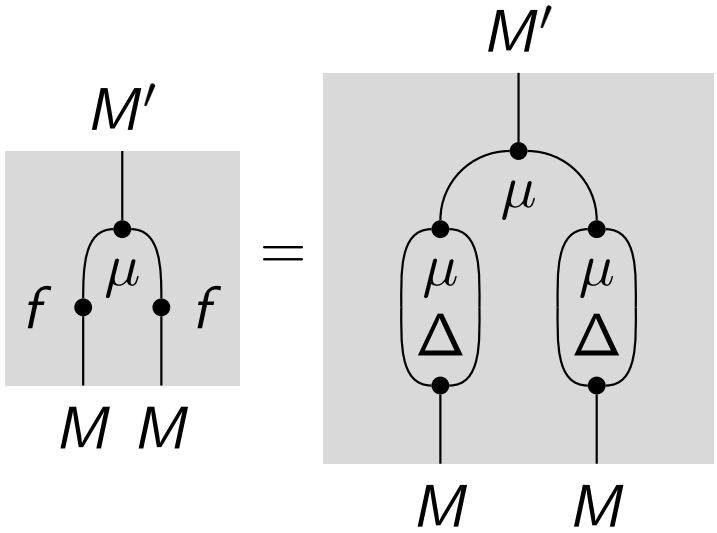
a homomorphism? (Equationally, $f(m) = m^2$.)

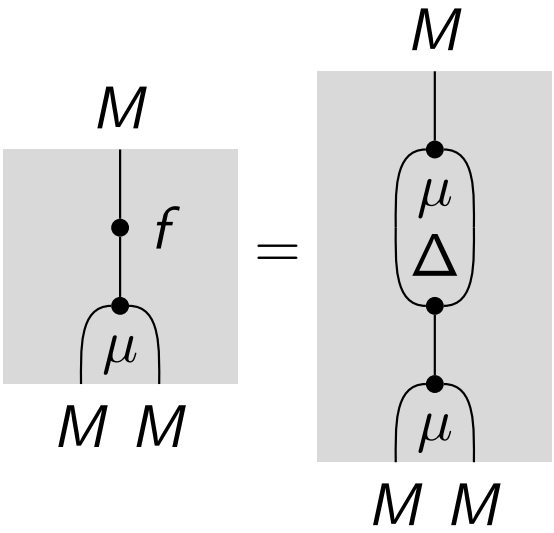
Is $f =$ 

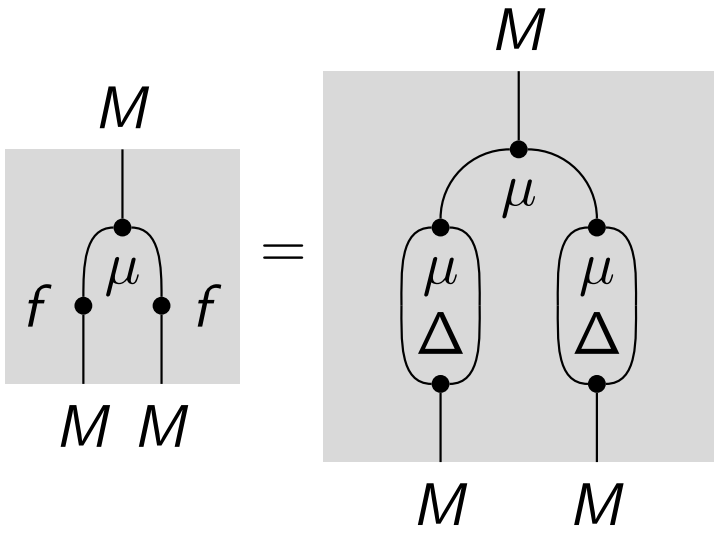
a homomorphism? (Equationally, $f(m) = m^2$.)

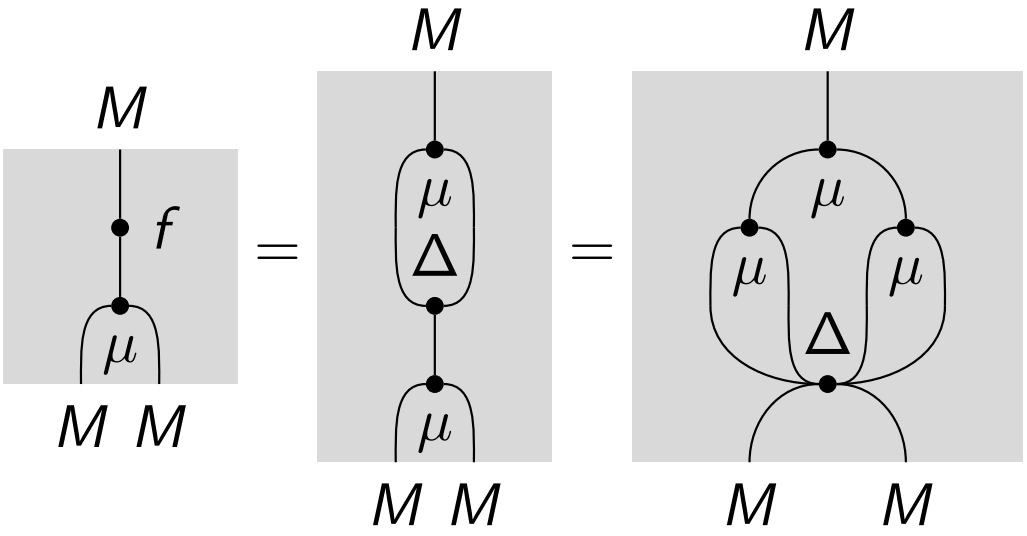
Well...

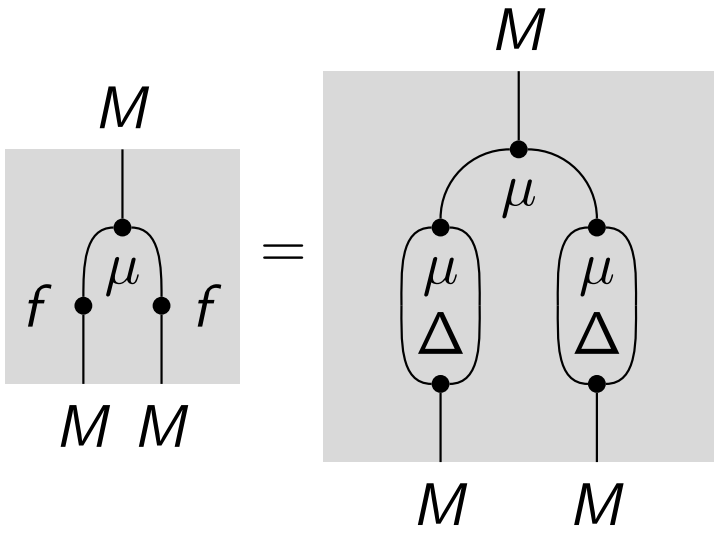


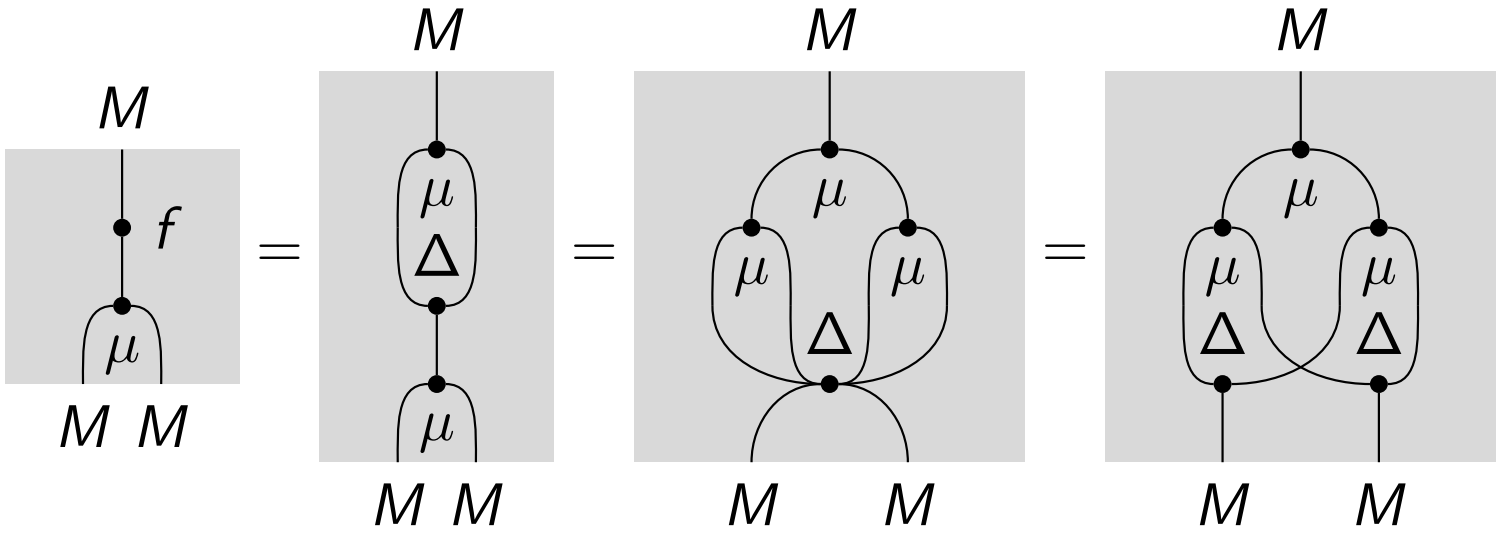


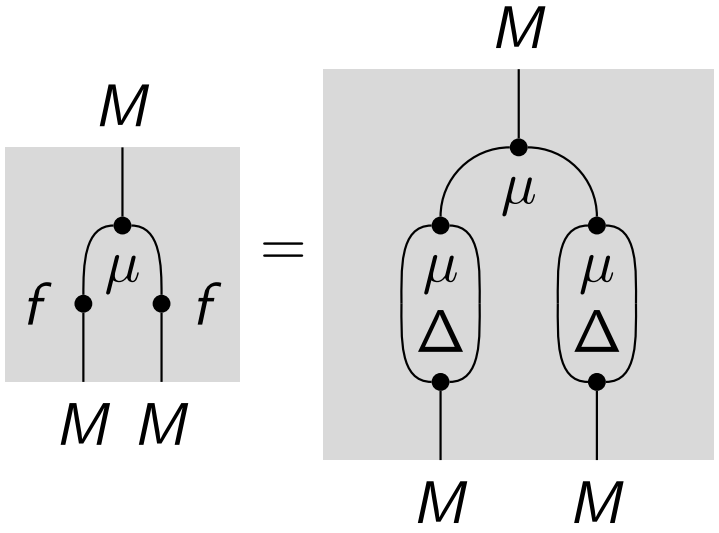

M
f
μ
M M
=
M
μ
Δ
μ
M M

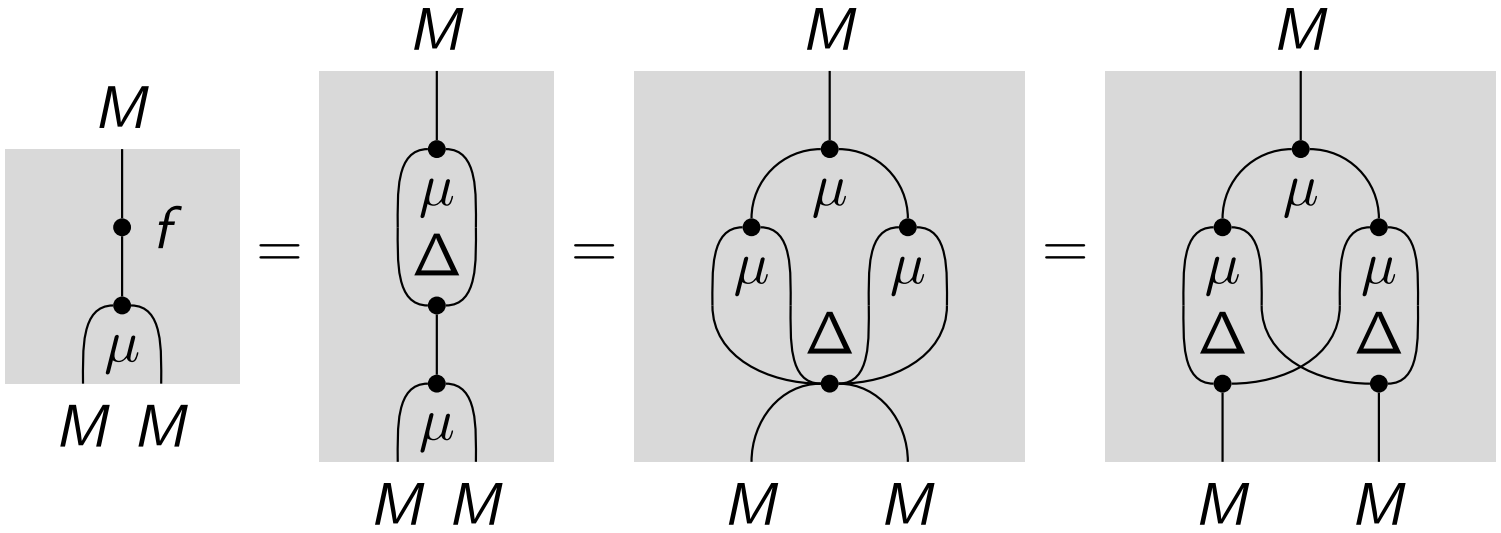

M
f
μ
f
M M
=
M
μ
μ
Δ
μ
Δ
M M

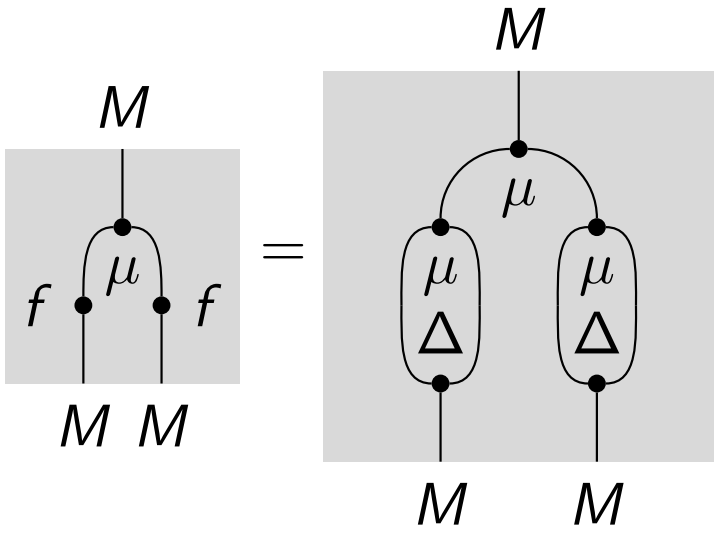

M
f
μ
M M
=
M
μ
Δ
μ
M M
=
M
μ
μ
μ
Δ
M M


M
f
μ
f
M M
=
M
μ
μ
Δ
μ
Δ
M M


M
f
μ
M M
=
M
μ
Δ
μ
M M
=
M
μ
μ
μ
Δ
M M
=
M
μ
μ
Δ
μ
Δ
M M


M
f
μ
f
M M
=
M
μ
μ
Δ
μ
Δ
M M





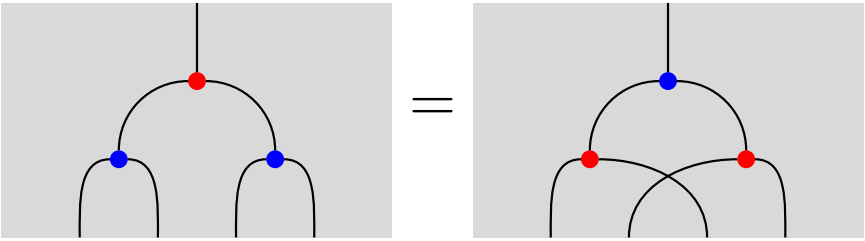
We need commutativity!

The Eckmann-Hilton argument

Let M be a set with two monoid structures $(M, \circ, e), (M, \otimes, e')$ so that

$$(a \otimes b) \circ (c \otimes d) = (a \circ c) \otimes (b \circ d).$$

Diagramatically,



Then $e = e'$, $\circ = \otimes$, and M is commutative.

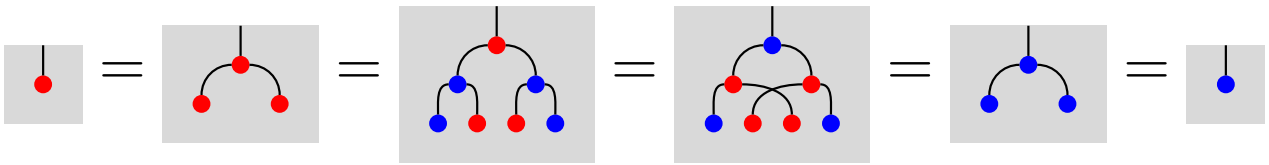
The Eckmann-Hilton argument

Proof.

$\square$

The Eckmann-Hilton argument

Proof.

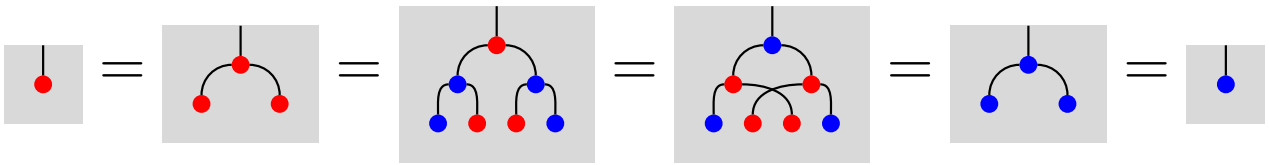


So $e = e'$.

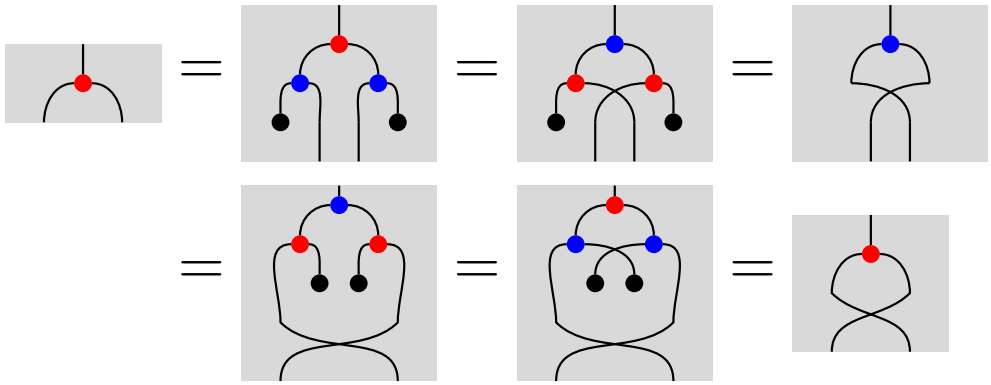
$\square$

The Eckmann-Hilton argument

Proof.



So $e = e'$.



So $b \otimes a = b \circ a$ and $a \otimes b = b \otimes a$ for all $a, b \in M$. $\square$